

Actuarial Decision-Support Prototype

****Prediction question.**** Given structured beneficiary-level utilization, cost, and demographic features from the current year, this system predicts whether the beneficiary will fall into the top decile of annual cost in the next year and then maps that risk into an operational intervention recommendation.

****Data source.**** The project uses the CMS DE-SynPUF synthetic Medicare claims data, with beneficiary summary, inpatient, outpatient, carrier, and prescription drug event files landed into volume-backed object storage and registered through a bronze, silver, and gold medallion pipeline in Databricks.

****Pipeline architecture.**** Raw CMS files are ingested into object storage, materialized into bronze Delta tables, standardized into silver entity tables, aggregated into a gold beneficiary-year feature table, used to train supervised risk models tracked in Databricks/MLflow, and then served through a FastAPI backend consumed by a Streamlit decision-support application.

****Model approach.**** Features are measured in the beneficiary feature year (t), while the label is computed from annual claim cost in target year ($t+1$). The top-decile threshold is computed only from training split. Logistic regression is retained as the interpretable statistical baseline. Gradient boosting is used as the primary operational model because it produced the strongest held-out ranking performance, with AUC-ROC = 0.8099, AUC-PR = 0.3879, and top-10 capture = 38.77%. XGBoost is in as a high-recall triage alternative rather than the preferred balanced model.

****What I learned.**** The hardest part was not fitting a classifier; it was maintaining a clean end-to-end contract across the pipeline. Bronze ingestion had to be hardened against CMS file irregularities, Unity Catalog restrictions changed how data could be staged, and the gold table design had to stay stable so that Databricks training, MLflow-tracked artifacts, the FastAPI schema, and the Streamlit UI all agreed on the same feature set. Class imbalance also mattered more than raw accuracy, so ranking metrics such as top-k capture and lift turned out to be more operationally meaningful than a single thresholded prediction score. The MDP/Q-learning layer is not trained from observed intervention outcomes; it is a simulated decision-support layer based on explicit transition and reward assumptions.